INCH-POUND

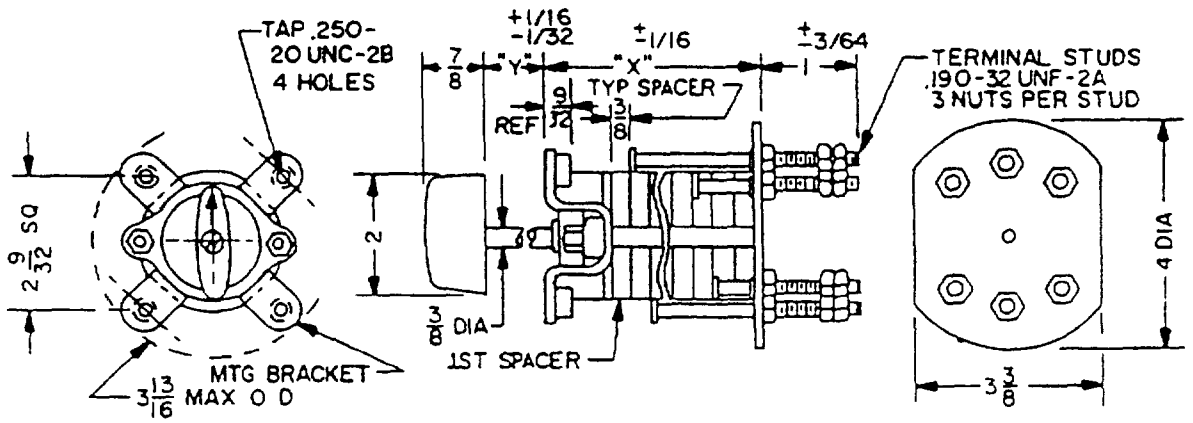
MIL-S-15291/5A(SH)
22 July 1991
SUPERCEDING
MIL-S-15291/5(SH)
17 April 1981

MILITARY SPECIFICATION SHEET

SWITCHES, ROTARY, SNAP ACTION CLASS 3SR FRONT MOUNTED, BACK CONNECTED

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-15291



NOTES
1 Dimensions are in inches
2 Unless otherwise specified, tolerance is plus or minus 0 015

FIGURE 1 *Class 3SR switch (up to 6 studs)*

AMSC N/A FSC 5930

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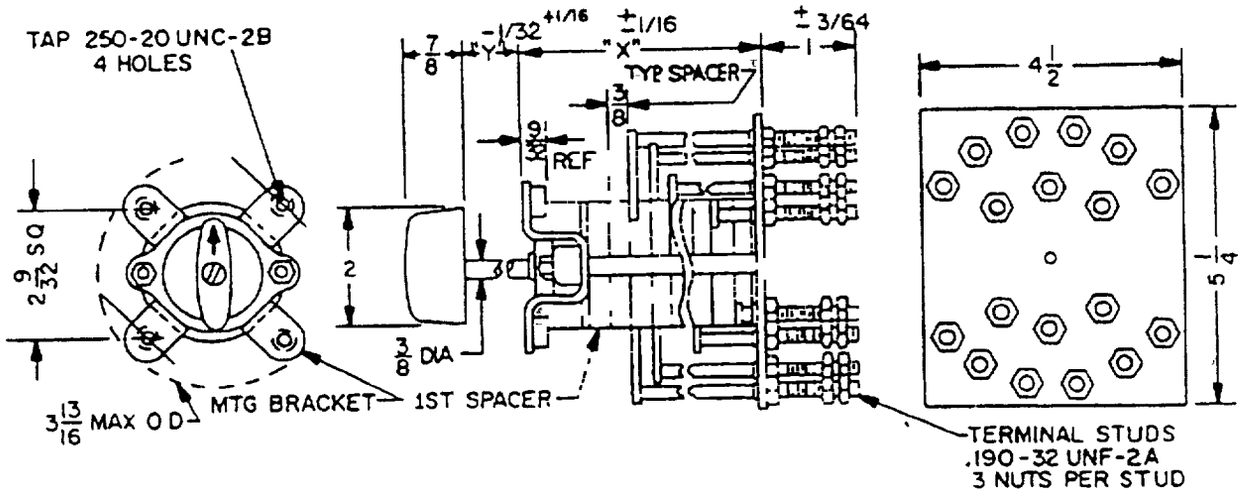
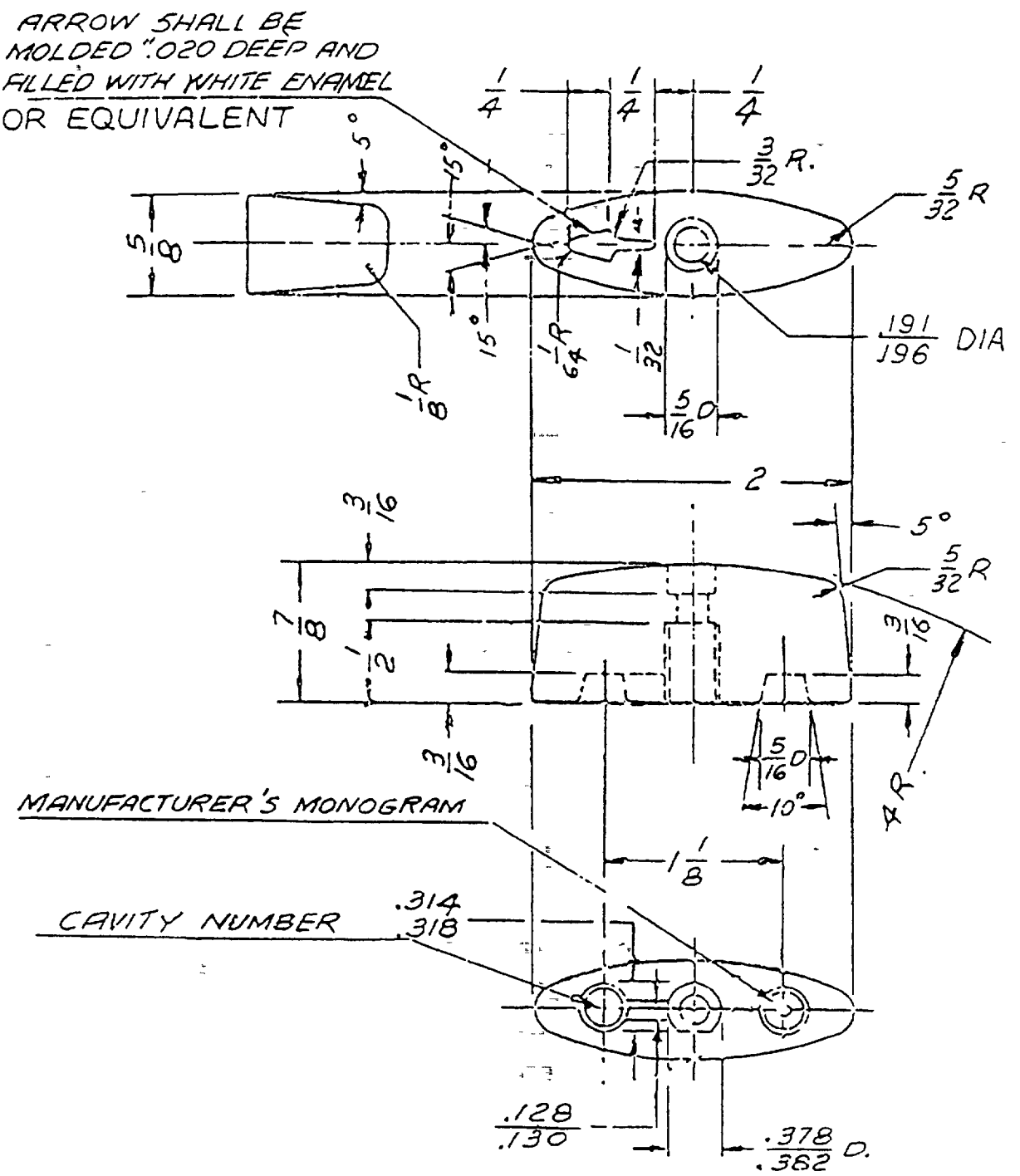


FIGURE 2. *Class 3SR switch (up to 18 studs)*

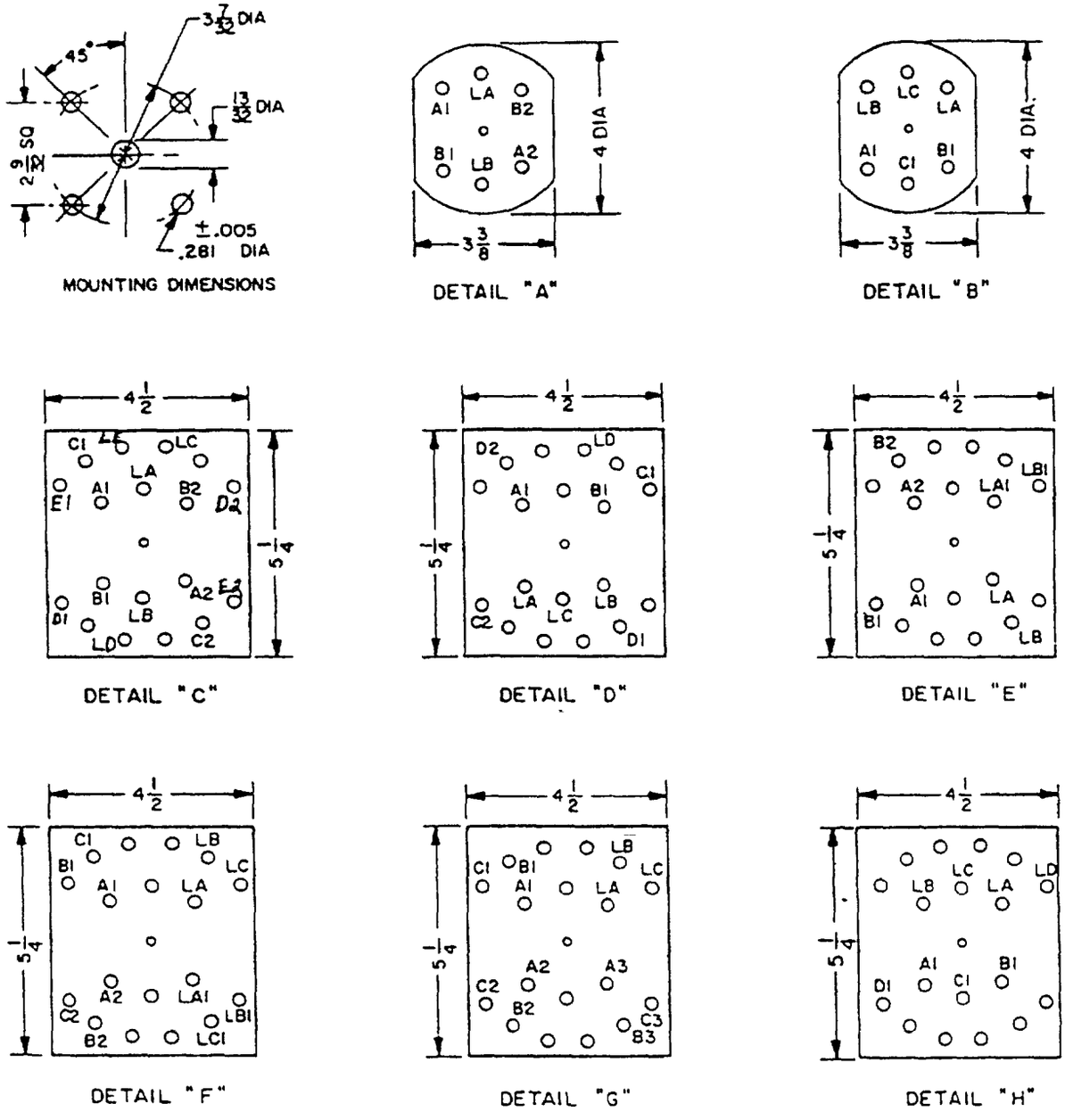
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PIN M15291/5-100

FIGURE 3. *Handle*

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NOTE
1 Viewed from the rear of the switch

FIGURE 4 *Mounting dimensions and terminal stud locations*

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TABLE I *Type and switching characteristics.*

MIL-S-15291/5 Dash No Type designation	Detail ref Torque in-lbs	Dim "X" Dim "Y"	Handle or shaft position	Circuit and spacer configuration rotor position, spacer locations, terminal marking and locations No 1	No 2	No 3	No 4	No 5	No 6	No 7	No 8	No 9	No 10	No 11	No 12	No 13	Notes
-001 3SR2A1	Fig 1 Detail B 9	2 3/8 7/16	OFF ON OFF		LB B	LA A1											
-002 3SR3A1	Fig 1 Detail B 9	2 3/4 7/16	OFF ON OFF		LC C1	LB B1	LA A1										
-003 3SR4A1	Fig 2 Detail H 9	3 1/8 7/16	OFF ON OFF		LD D1	LC C1	LB B1	LA A1									
-004 3SR2C1	Fig 1 Detail A 9	3 1/8 7/16	OFF ON 1 2 OFF		LC LB	B2 B1	LA A2	A1									
-005 3SR3B1	Fig 2 Detail C 9	3 7/8 7/16	OFF ON 1 2 OFF		LC C2	C1 C2	B2 B1	LB	LA	A1 A2							
-006	Fig 2 Detail C 9	5 3/8 7/16	OFF ON 1 2 OFF		LF E2	E1 E2	D2 D1	LD	LC	C1 C2	B2 B1	LB	LA	A1 A2			
-007 3SR3E1	Fig 2 Detail G 9	3 7/8 7/16	OFF ON 1 3 ON 2		LC	C1 C3	L B1 B2 C2	B1 B3 B2	LA	A1 A3 A2							
-008 3SR4F1	Fig 2 Detail E 9	3 1/8 7/16	OFF ON ON 1&2		LB B1	B2	LA A1	LA1 A2									
-009 3SR6F1	Fig 2 Detail F 9	3 7/8 7/16	OFF ON 1 2 ON 1&2		LC1 C2	LC C1	LB1 B2	LB B1	LA1 A2	LA A1							
-010 3SR4S1	Fig 2 Detail D 9	3 7/8 7/16	OFF ON OFF		LD	D2 D1	LC0	C1 C2	LA1 B1	LA LB							-1

[1]010 Switch was listed incorrectly as 3SR4B1 in MIL-S-15291C (SHIPS)

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REQUIREMENTS:

Applicable specification.	MIL-S-15291
Dimensions and mounting	See figures 1 through 4
Switching characteristics	See table I
Angle of throw.	90 degrees
Switching action.	Snap action, reciprocating
Electrical and endurance ratings	See table II
Stop strength (applicable to switches with stops)·	100 inch-pounds
Vibration·	50 Hz, MIL-STD-167-1
Shock:	High impact, MIL-S-901
Insulation (spacers and decks)	Glass alkyd, MAI-60 (or MAI-30) in accordance with MIL-M-14
Overload:	600 percent of ac rating, 50 make and break operations
Contact resistance:	0.01 ohms maximum
Dielectric withstanding voltage	2,000 Vrms
Insulation resistance:	200 megohms minimum
Temperature rise	50 °C maximum
Contact material.	Rotor, phosphor bronze – Stationary contact, copper
Mounting screws.	.250-20UNC-2A (4) length and headstyle to suit application. Screws not furnished.

TABLE II *Electrical and endurance ratings*

Tests	Current (amperes)	Voltage (volts)	Electrical operations (number of operations)	Test rate (operations per minute)	De-energized operations (number of operations)
Alternating current (rms)	30	500	20,000	30	10,000
Direct current	30	250	15,000	15	-

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Terminal marking

Terminal markings shall be in accordance with details A through H, as specified in table I Terminal markings in table I are as viewed from the front (handle end) of the switch

GENERAL INFORMATION:

Switches not covered by specification sheets:

Switches which are fabricated from standard parts, as used in qualified switches, but which do not comply with switches detailed herein with respect to circuit characteristics, switching action, mounting arrangement, and handle details may be acquired under this specification from contractors having qualification approval under this specification.

Extended ratings:

Switches detailed herein have been tested and found satisfactory at the extended ratings listed in table III. Reduced life expectancy must be anticipated for switches used at these increased voltage or current levels. Tests under the conditions of table III are not required for qualification acceptance and they are not repeated routinely as for maintenance of qualification. Supplemental evaluations and tests applicable to particular circuit requirements are recommended.

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TABLE III *Extended ratings.*

Switching characteristics	AC - 60 or 400 Hz											
	125 volts				250 volts				500 volts			
	Resistive or lamp load		Inductive load 0.75 p f		Resistive or lamp load		Inductive load 0.75 p f		Resistive or lamp load		Inductive load 0.75 p f	
	Amp	Operations	Amp	Operations	Amp	Operations	Amp.	Operations	Amp	Operations	Amp	Operations
All	30	20,000	30	20,000	30	20,000	30	20,000	30	20,000	30	20,000
E, F	40	6,000	40	6,000	---	---	---	---	---	---	---	---
A B [1]	40	6,000	40	6,000	30	6,000	30	6,000	30	6,000	30	6,000
	DC											
	120 volts				250 volts				350 volts			
	Resistive or lamp load		Inductive load [2]		Resistive or lamp load		Inductive load [2]		Resistive or lamp load		Inductive load [2]	
	Amp	Operations	Amp.	Operations	Amp.	Operations	Amp	Operations	Amp.	Operations	Amp	Operations
All	30	13,000	25	6,000	15	13,000	---	---	---	---	---	---
A, B	30	15,000	25	6,000	25	6,000	25	6,000	15	6,000	15	6,000
A, B [3]	---	---	---	---	30	15,000	---	---	---	---	---	---

[1] Single pole break – all other ratings are based on breaking both sides of the line in accordance with figures 1 and 2 of MIL-S-15291

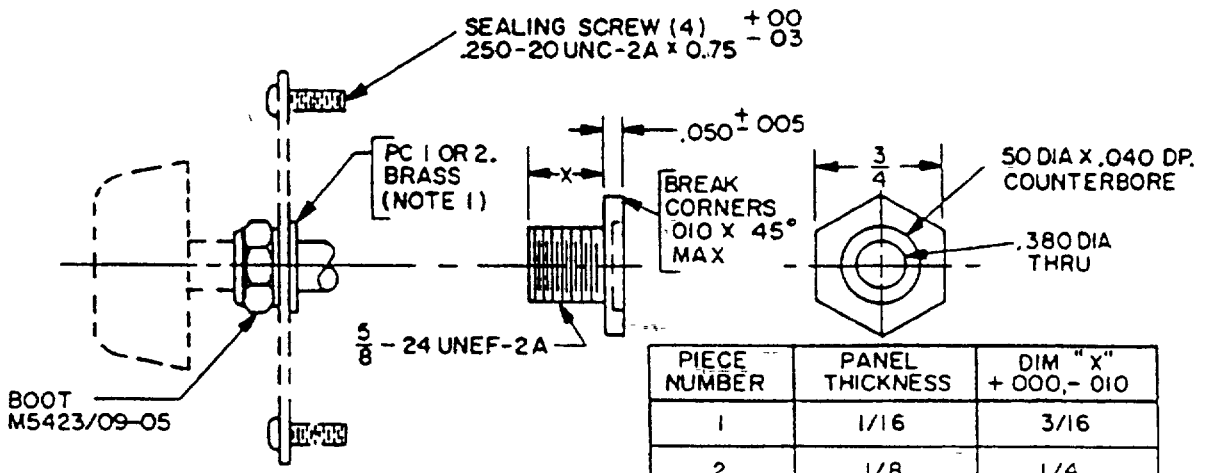
[2] 0.08 henry for inductive circuit

[3] Switch with special arc snuffers

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Sealed mounting

Switches listed in table I as having "Y" dimension not less than 7/16 inch can be mounted with a kit, figure 5, to effect a submersible seal (SBM 15) between switch and mounting panel. Panel drilling must comply with figure 6. Kits not supplied unless specified (see 6.2 of MIL-S-15291).



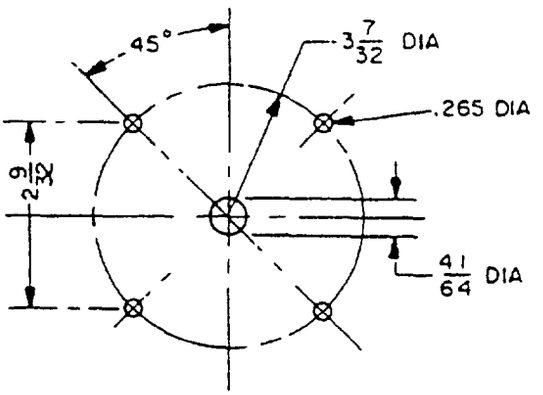
PIECE NUMBER	PANEL THICKNESS	DIM "X" +.000,-.010
1	1/16	3/16
2	1/8	1/4

NOTE

1 Finish shall be zinc and dichromate, ASTM B 633

FIGURE 5 *Kit for panel seal mounting*

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NOTE.
1 Do not chamfer mounting holes Breaking 1/64 maximum permissible

FIGURE 6 *Panel drilling for mounting kit*

APPLICATION AND ACQUISITION GUIDE: PIN and type designation cross reference shall be as shown in table IV.

TABLE IV. *Application and acquisition guide*

M15291/5 dash No	Type designation	For new or existing design	For re-placement	Circuit configuration
-001	1SR2A1	X		A (off-on-off-on)
-002	1SR3A1	X		A (off-on-off-on)
-003	3SR4A1	X		A (off-on-off-on)
-004	3SR2B1	X		B (off-on 1-oft-on 2)
-005	3SR3B1	X		B (off-on 1-off-on 2)
-006	–	X		B (off-on 1-off-on 2)
-007	3SR3E1	X		E (off-on 1-on 2-on 3)
-008	3SR4F1	X		F (oft-on 1-on 1 & 2-on 2)
-009	3SR6F1	X		F (off-on 1-on 1 & 2-on 2)
-010	3SR4S1	X		S (see table I)

Revision letters are not used to denote changes due to the extensiveness of the changes

Preparing activity.
Navy - SH
(Project 5930-N663-6)

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